

Amjid Ali

Computer Vision Engineer | Real-Time Video Analytics & Industrial Safety AI

amjidali3797@gmail.com | +82 10 2158 1412 | D-10 Visa | Gwacheon, Korea

[Github](#) | [LinkedIn](#) | [Google Scholar](#)

서울특별시 광진구 군자동 60-57

SUMMARY

Computer Vision Engineer building real-time video analytics systems that run in production on live industrial CCTV. Currently delivering safety detection AI for POSCO's Gwangyang steel plant and two other Korean manufacturers, covering fire and smoke, worker safety, and heavy-equipment collision detection across continuous camera streams. Experienced end-to-end: dataset construction, model training, inference optimization, and deployment as GPU video pipelines. M.S. in Software with published research in audio-visual anomaly recognition, including IEEE Access. Hands-on experience with PyTorch, TensorFlow, YOLO, DeepStream, OpenCV, FFmpeg, MediaMTX, FastAPI, and edge deployment.

SKILLS

▶ TensorFlow	▶ PyTorch	▶ Keras	▶ Transformers	▶ CNN	▶ RNNs	▶ OpenCV
▶ Object Segmentation & Detection	▶ Multimodal Fusion			▶ Anomaly Detection	▶ Edge Deployment	
▶ Video Analytics	▶ Git	▶ LangChain		▶ RAG	▶ LLM Fine-tuning	

EXPERIENCE

VoltaON AI Solution, Gwacheon, SOUTH KOREA

FEBRUARY 2026 - Current

AI Developer

Projects:

- **Industrial Safety AI System POSCO, Gwangyang**
 - Built a real-time video monitoring and safety detection system for POSCO's Gwangyang steel plant, using computer vision and deep learning to identify critical industrial safety events as they occur.
 - Delivered detection modules for fire and smoke, worker safety monitoring, equipment safety, and heavy-equipment collision detection.
 - Built computer vision pipelines that detect and track workers, vehicles, and industrial equipment across live CCTV streams.
 - Engineered real-time inference and video processing pipelines with DeepStream, PyTorch, YOLO, OpenCV, FFmpeg, and MediaMTX, served through FastAPI.
- **Fire Detection AI System Seyoung**
 - Delivered a real-time fire detection system using computer vision and deep learning for automated fire monitoring.
 - Built and integrated video-based fire detection pipelines to identify potential fire incidents from surveillance footage.
- **Industrial Safety AI System Korea THT**
 - Developed heavy-equipment collision and fire detection systems using computer vision for real-time industrial safety monitoring.
 - Built video-based detection pipelines to identify vehicle collisions and fire incidents from surveillance footage.

SEJONG UNIVERSITY, SEOUL, SOUTH KOREA

MARCH 2024 - FEBRUARY 2026

Research Assistant – Intelligent Media Laboratory

Projects:

- **RAG Document Intelligence Chatbot**
 - Developed a smart chat that allows users to "chat" with their documents to find exact information without manual reading.
 - Improved system accuracy by structuring documents into logical sections for reliable, context-aware information retrieval.
- **Development of Advanced algorithm for battery health prediction**
 - Developed models that analyze battery data to estimate state of health, performance, and remaining useful life, enabling early failure prediction.

- Used deep learning models for data imputation and sequential models for prediction.
- **AI policy simulation platform to solve social problems in depopulated areas in Korea**
 - Built an AI-based policy simulation platform to analyze and predict the impact of government policies on depopulation in rural areas of Korea.
 - Trained and compared multiple machine learning models, handled data processing, and implemented forward prediction.
- **Unity Scenario Development**
 - Developed Unity-based simulated environments to generate high-quality synthetic data for AI and computer vision research.
 - Designed virtual scenes and scenarios, controlled lighting, camera angles, and object behaviors, and generated datasets for training and testing machine learning models.
- **Anomaly Behavior recognition for accident prevention in a connected vision environment (Thesis Work)**
 - Developed an intelligent surveillance system that detects abnormal or dangerous situations in real time using multimodal visual and audio data.
 - Implemented computer vision and deep learning models, integrated audio-visual fusion, and built a benchmark dataset for anomaly detection of accidents and unsafe behaviors.

ISLAMIA COLLEGE PESHAWAR UNIVERSITY

January 2020 -January 2023

Research Assistant – Digital Image Processing laboratory

Projects:

- **Early Fall Detection**
 - Developed an AI-based early fall detection system that analyzes gait and movement patterns to identify fall risks in elderly individuals, supporting early intervention and improved safety.
 - Applied machine learning to spatiotemporal motion data, modeled gait irregularities, developed user interfaces, and contributed to an EU-funded smart healthcare research initiative for aging populations.
- **Face Recognition**
 - Developed a real-time face recognition system deployed on a Raspberry Pi for lightweight and edge-based identity recognition.
 - Implemented and deployed computer vision and face recognition models on edge device, including camera integration and real-time inference under hardware constraints.

EDUCATION

Sejong University, Seoul, South Korea

March 2024 -February 2026

Master's in software - Intelligent Media Laboratory

Research Publications:

- AVAR-Net: A Lightweight Audio-Visual Anomaly Recognition Framework with a Benchmark Dataset (Under review)
- Dual-Modality Visual-Audio Fusion and a New Benchmark for Anomaly Recognition in Dynamic Scenes (Submitted)
- Multi Camera Connected Vision System with Multi View Analytics: A comprehensive Survey [[arXiv](#)]
- Active Learning for Anomaly Recognition: Leveraging Visual and Audio Data Fusion
- Audio-Visual Semi-Supervised Framework for Real-Time Anomaly Recognition

Ads Modified ViT-based Anomaly Recognition using Audio Data

Islamia College Peshawar University, Pakistan

September 2018 -September 2022

Bachelor's in software engineering – Software Development

Research Publications:

- [Abandoned object detection and classification using deep embedded vision](#), IEEE Access (2024).

Projects:

- **Visually Assisted System Using Blockchain (Thesis)**
 - Developed a secure computer vision framework that combines deep learning-based visual analysis with

blockchain technology to ensure transparency and data integrity.
Integrated Ethereum smart contracts for tamper-proof storage and verification and implemented deep learning models for video summarization and visual data extraction.

RESEARCH & PUBLICATIONS

- AVAR-Net: A Lightweight Audio-Visual Anomaly Recognition Framework with a Benchmark Dataset (Under Review)
- Dual-Modality Visual-Audio Fusion and a New Benchmark for Anomaly Recognition in Dynamic Scenes (Submitted)
- Multi Camera Connected Vision System with Multi View Analytics: A Comprehensive Survey (arXiv)
- Active Learning for Anomaly Recognition: Leveraging Visual and Audio Data Fusion
- Audio-Visual Semi-Supervised Framework for Real-Time Anomaly Recognition
- Modified ViT-based Anomaly Recognition using Audio Data • Abandoned Object Detection and Classification Using Deep Embedded Vision (IEEE Access)

ACHIEVEMENTS & AWARDS

Prime Minister Youth Initiative Scheme	March 2021
<i>Awarded Laptop by Prime Minister for educational performance</i>	
100% Study Scholarship	December 2023
<i>Sejong University, South Korea</i>	
AI for Material Science	October 2024
<i>Awarded Cash Prize by the CEO of IM Lab for the performance</i>	